

From the SSDs, we can clearly identify three critical system operations namely:

1. createTask(...)
2. updateTask(...)
3. viewTask(...)

For these system operations we can create the following operation contracts:

Contract CO1 :	Create Task
Operation :	createTask(String title, String description)
Cross References :	Use Case: Create Task
Preconditions :	• Title must not be empty
Postconditions :	• A new instance t:Task is created • t is added to TaskDB • A new instance r:Record is created • r is added to RecordDB

Contract CO2 :	Update Task
Operation :	updateTask(Task t, String type, String attribute)
Cross References :	Use Case: Update Task
Preconditions :	• Task must exist in the TaskDB • $type \in \{ \text{"editTaskTitle", "editTaskDescription", "setPriority", "setDueDate", "setStatus", "addTask", "removeTask", "moveTask", "addTag" } \}$
Postconditions :	• If type is "editTaskTitle", t.title is updated to attribute • If type is "editTaskDescription", t.description is updated to attribute • If type is "setPriority", t.priority is updated to attribute • If type is "setDueDate", t.dueDate is updated to attribute • If type is "setStatus", t.status is updated to attribute • If type is "addTask", t is associated with a Project • If type is "removeTask", t is disassociated from its Project • If type is "moveTask", t is removed from its current Project

and t is associated with another Project (the target project must exist)

- If type is “addTag”, a tag is added to t
- A new instance r: Record is created
- r.description reflects the update action
- r is stored in RecordDB

Contract CO3 :

View Task

Operation :

viewTask(String[] keywords, String type)

Cross References :

Use Case: View Task

Preconditions :

- keywords array must not be empty
- OR
- $\text{type} \in \{ \text{“byDueDate”}, \text{“byPriority”}, \text{“byStatus”}, \text{“byProject”}, \text{“byTag”} \}$

Postconditions :

- If keywords array is not empty AND type is empty, tasks whose title or description contain keywords are retrieved from TaskDB
- If keywords array is not empty AND type is not empty, tasks whose title or description contain keywords are retrieved from TaskDB and regrouped according to the type
- If keywords array is empty AND type is “byDueDate”, tasks are returned ordered/grouped by due date
- If keywords array is empty AND type is “byPriority”, tasks are returned ordered/grouped by priority
- If keywords array is empty AND type is “byStatus”, tasks are returned grouped by status
- If keywords array is empty AND type is “byProject”, tasks are returned grouped by project
- If keywords array is empty AND type is “byTag”, tasks are returned grouped by tag